

LAWRENCE, KS, USA

WMO: 724508

Lat: 39.008N

Lon: 95.212W

Elev: 254

StdP: 98.32

Time Zone: -6.00 (NAC)

Period: 97-19

WBAN: 03997

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.0	-13.0	-20.6	0.6	-14.8	-18.0	0.8	-11.6	12.4	1.9	11.3	1.2	2.8	320	0.524

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.9	37.1	24.8	35.0	24.5	33.2	24.2	26.6	33.5	25.8	32.5	25.1	31.6	4.9	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.7	20.3	30.5	23.9	19.3	29.7	23.0	18.3	28.7	84.5	33.6	81.1	32.8	77.9	31.6	29.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.2	9.8	8.6	-21.2	39.6	3.5	1.9	-23.7	41.0	-25.7	42.1	-27.6	43.2	-30.1	44.6		
(5)			-21.5	28.1	3.3	0.8	-23.9	28.7	-25.8	29.2	-27.7	29.6	-30.1	30.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	13.0	-1.2	1.1	7.2	12.9	18.6	23.9	26.3	25.3	20.7	13.6	6.8	0.7		
	DBStd	10.87	6.16	6.35	6.25	5.28	4.64	3.53	3.22	3.43	4.49	5.34	5.58	5.93		
	HDD10.0	1207	348	255	129	30	2	0	0	0	0	25	126	293		
	HDD18.3	2757	606	482	349	175	56	3	0	1	26	166	347	547		
	CDD10.0	2318	2	6	41	117	269	416	504	474	322	135	29	4		
	CDD18.3	826	0	0	3	12	64	169	246	216	97	17	1	0		
	CDH23.3	8964	0	1	29	146	636	1796	2826	2323	1023	177	7	0		
	CDH26.7	4064	0	0	4	31	204	768	1429	1154	424	48	1	0		
(14) Wind	WSAvg	3.8	3.9	4.1	4.5	4.8	4.0	3.6	2.9	2.8	3.1	3.6	3.9	3.8		
(15) Precipitation	PrecAvg	999	30	31	64	92	126	152	109	102	115	83	60	42		
	PrecMax	1505	89	80	268	227	293	413	465	278	326	257	158	121		
	PrecMin	548	0	0	11	17	18	52	6	16	14	5	0	0		
	PrecStd	244	22	20	46	55	63	93	86	65	82	56	42	31		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.0	22.0	26.7	30.0	32.9	36.9	39.1	39.5	36.2	31.0	24.1	19.1		
		MCWB	10.2	13.4	17.1	19.1	22.6	24.6	24.2	24.2	23.2	21.4	16.5	14.2		
	2%	DB	14.2	17.8	23.1	27.1	31.0	34.2	37.2	37.1	33.0	27.7	21.9	15.9		
		MCWB	8.6	11.8	15.1	17.9	21.6	24.3	25.2	24.6	22.8	19.1	15.4	11.5		
	5%	DB	11.4	14.4	20.9	24.8	28.9	32.7	35.2	34.7	31.3	24.9	18.9	12.6		
		MCWB	6.7	9.2	13.9	16.8	20.8	23.9	25.2	24.3	22.9	17.8	13.1	8.8		
	10%	DB	8.1	11.4	17.9	22.5	27.1	31.1	33.2	32.6	28.9	22.7	16.3	10.0		
		MCWB	4.5	7.0	12.3	15.9	20.1	23.4	24.8	24.1	21.3	17.0	11.6	6.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.0	15.4	18.8	20.8	24.3	26.7	27.9	27.2	25.5	22.9	18.1	15.7		
		MCDB	15.6	19.0	24.3	27.5	30.6	33.1	34.2	34.0	31.8	28.4	21.9	18.4		
	2%	WB	9.3	12.3	17.0	19.5	23.0	25.6	26.8	26.2	24.3	21.4	15.8	12.3		
		MCDB	13.3	16.5	21.3	24.7	28.7	32.2	33.7	33.5	30.8	25.7	20.0	14.7		
	5%	WB	6.9	9.8	15.2	18.2	21.9	24.8	26.1	25.4	23.5	19.6	14.3	9.0		
		MCDB	11.0	14.7	19.3	23.2	27.3	31.1	32.9	32.4	29.4	23.2	17.7	12.0		
	10%	WB	4.7	7.1	12.6	16.6	20.8	23.9	25.4	24.6	22.5	17.8	12.2	6.7		
		MCDB	8.4	11.2	17.3	21.9	25.7	29.7	32.0	30.9	27.8	21.4	15.9	9.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.8	12.0	12.8	13.0	12.2	11.7	11.9	12.5	13.5	13.3	12.8	11.4		
		MCDBR	17.1	17.5	17.2	16.2	14.1	13.0	14.1	15.1	14.5	15.4	16.2	15.9		
		MCWBR	11.9	11.7	10.2	9.0	6.8	5.5	5.1	5.4	6.1	8.3	10.2	11.4		
	5% WB	MCDBR	15.8	16.1	13.4	13.5	12.0	11.8	11.9	12.8	12.6	11.9	13.5	14.0		
		MCWBR	11.4	11.5	8.7	8.5	6.6	5.8	5.5	5.7	5.9	7.6	9.7	11.1		
(40) Clear-Sky Solar Irradiance	taub		0.292	0.305	0.335	0.388	0.425	0.434	0.438	0.438	0.389	0.343	0.308	0.287		
	taud		2.474	2.447	2.422	2.292	2.246	2.251	2.266	2.272	2.358	2.475	2.515	2.510		
	Ebn at Noon		890	923	924	888	858	848	840	831	854	867	863	870		
	Edh at Noon		79	93	106	128	137	137	134	130	112	90	75	71		
(44) All-Sky Solar Radiation	RadAvg		2.24	3.03	3.92	5.00	5.73	6.50	6.46	5.75	4.80	3.45	2.45	1.93		
	RadStd		0.22	0.34	0.33	0.32	0.50	0.46	0.42	0.29	0.33	0.39	0.23	0.18		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	(e) -0.54	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
(47)	Regional (26 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page